A LOCALLY FINITE TYPE MONOMORPHISM OF ALGEBRAIC SPACES IS SEPARATED AND LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof reduces to the affine base and then works with the identity U x_X U = U x_Y U coming from the monomorphism hypothesis; that identity is what makes the fibres finite.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A monomorphism of algebraic spaces is separated. Lemma lemma-monomorphism-separated, used at proof lines 5586-5587. Gives the separatedness half of the conclusion.
- Local quasi-finiteness is local on the base. Lemma lemma-quasi-finite-local, used at proof lines 5588-5591. Reduces to the case of an affine base Y.
- Fibres of etale morphisms of schemes over a field are finite discrete. Morphisms, Lemma morphisms-lemma-etale-over-field, used at proof lines 5600-5603. Shows the fibres of U -> Y are finite.
- The criterion for quasi-finiteness at a point via finite fibres. Morphisms, Lemma morphisms-lemma-quasi-finite-at-point-characterize, used at proof lines 5604-5605. Converts finiteness of the fibres into quasi-finiteness of U -> Y.

Source reference. Original lemma 042N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5334–5334. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 040Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5274–5274. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3232–3232. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01TH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2374–2374. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. If f is locally of finite type and a monomorphism, then f is separated and locally quasi-finite.*

Proof. A monomorphism is separated, see Lemma 042N. By Lemma 040Z it suffices to prove the lemma after performing a base change by $Z \to Y$ with Z affine. Hence we may assume that Y is an affine scheme. Choose an affine scheme U and an étale morphism $U \to X$. Since $X \to Y$ is locally of finite type the morphism of affine schemes $U \to Y$ is of finite type. Since $X \to Y$ is a monomorphism we have $U \times_X U = U \times_Y U$. In particular the maps $U \times_Y U \to U$ are étale. Let $y \in Y$. Then either U_y is empty, or $\mathrm{Spec}(\kappa(u)) \times_{\mathrm{Spec}(\kappa(y))} U_y$ is isomorphic to the fibre of $U \times_Y U \to U$ over u for some $u \in U$ lying over y. This implies that the fibres of $U \to Y$ are finite discrete sets (as $U \times_Y U \to U$ is an étale morphism of affine schemes, see Morphisms, Lemma 02GL). Hence $U \to Y$ is quasi-finite, see Morphisms, Lemma 01TH. As $U \to X$ was an arbitrary étale morphism with U affine this implies that $X \to Y$ is locally quasi-finite. □